

Feasibility-Aware Benchmarking of a HUBO Formulation for Occlusion-Aware UAV Grid Path Planning

Abstract—A vehicle navigating a cluttered environment toward a goal must avoid obstacles while keeping that goal visually unoccluded. Each subproblem has been extensively studied; a common approach to the joint problem uses continuous Model Predictive Control (MPC). This paper asks whether the same problem class can be expressed as a discrete higher-order binary optimization problem, and what a carefully controlled comparison against classical baselines shows.

We contribute: (1) a Higher-Order Unconstrained Binary Optimization (HUBO) formulation of occlusion-aware grid path planning whose cubic term penalizes three consecutive steps within a connected chain of near-obstacle buffer cells; (2) a *feasibility-aware* evaluation protocol that separates structural feasibility, goal arrival, and objective quality, verifies every constraint by direct inspection rather than energy thresholds, and characterizes penalty-weight exploitation via a sweep; and (3) a measured account of the cost of reducing this HUBO to QUBO form. On an 8×8 instance with $L=15$ time layers (960 binary variables; 3,172 cubic monomials), `dimod` quadratization adds 582 auxiliary variables (+60 %), giving a 1,542-variable QUBO with 88,470 quadratic interactions—which is what `neal`, a classical simulated annealing sampler, in fact receives.

Under a single external evaluator, a classical A* baseline and a constrained trajectory-space SA attain soft costs of 317 and 303 with 50/50 structurally feasible samples, though only 7/50 SA samples reach the target; `neal` on the reduced QUBO attains 11/50 (22%, 95% CI 12.8–35.2%) structural feasibility at $\lambda=1000$. A separate 15-read sweep gives 0/15 feasible at $\lambda \leq 300$ rising to 9/15 at $\lambda=2000$. We report the `dimod` reduction penalty actually used (the default, 1.0) and the resulting reduction-violation rate, and we identify this as a confound that must be controlled before attributing `neal`’s behaviour to quadratization. Code: [\[repository link\]](#).

Index Terms—HUBO, QUBO, quadratization, simulated annealing, QAOA, occlusion-aware planning, grid path planning, A*, feasibility-aware benchmarking.

I. INTRODUCTION

A. Motivation

Consider a quadrotor following a person across an urban plaza dotted with kiosks and pillars. It must not collide with an obstacle, and it must keep the person in camera view so the tracker does not lose lock. Each subproblem has been extensively studied; what makes the joint problem interesting is that the two objectives trade off. A path that hugs an obstacle for cover is short and collision-free, but if that obstacle blocks the line of sight for several seconds, tracking degrades. A path arcing around obstacles to preserve sight lines may exceed the vehicle’s flight time.

A common approach for continuous control folds both costs into an MPC objective solved over a receding horizon; Masnavi *et al.* [6] formulate a related continuous 3D occlusion-free tracking problem as a real-time multi-convex MPC. We instead study a *discretized* abstraction of the same problem class: a 2D grid path-planning model in which the decision is which cell the vehicle occupies at each time layer. We emphasize at the outset that this is a grid path-planning abstraction: it has no velocity, acceleration, heading, camera dynamics, or control inputs, and results here do not transfer directly to a full quadrotor trajectory optimizer.

B. Why a Higher-Order Model

Discretizing lets us write the problem as a binary polynomial: $x_{t,v} \in \{0,1\}$ is 1 iff the vehicle occupies cell v at layer t . Structural constraints and costs become polynomial in these variables. Most terms are linear or quadratic. One is not: we penalize the vehicle for spending *three consecutive layers* inside the set of near-obstacle “buffer” cells, which produces a genuine degree-three monomial (Section III-E). Reducing that term to quadratic form requires auxiliary variables via Rosenberg reduction [10], [11], at a cost we measure directly.

C. Scope of Claims

This paper does not claim quantum speedup. The instances are small, all full-size runs are classical, and the QAOA component is a simulator-scale encoding check. We claim a discrete HUBO formulation of an occlusion-aware grid planning problem, an evaluation protocol that prevents a class of misleading comparisons, and a measured account of quadratization cost. We identify explicitly, in Section VI, the confounds that prevent a causal claim about *why* the QUBO-side sampler underperforms.

II. RELATED WORK

Visibility-aware UAV target tracking has been addressed in continuous form by Chen *et al.* [1], Hausman *et al.* [2] (occlusion-aware cooperative 3D tracking), Penin *et al.* [3] (collision and occlusion avoidance), Falanga *et al.* [4] (perception-aware MPC, targeting point-of-interest visibility rather than moving-target occlusion specifically), and Wang *et al.* [5] (multi-UAV tracking with maximized visibility). Masnavi *et al.* [6] address a related continuous 3D occlusion-free tracking problem; our discrete model shares its visibility motivation but not its dynamics or tracking objective.

In quantum optimization, QUAU [7] solves single-UAV obstacle-avoidance path planning with QAOA; Q4DR [8] addresses drone *routing* through hybrid annealing and gate-based decomposition. Adjacent work includes path planning for critical ATM/UTM corridors using quantum and high-performance computing [16], QUBO/QAOA drone mission planning [17], hybrid quantum-classical next-best-view planning [18], and hybrid quantum-classical multi-agent pathfinding via iterated QUBO [19]. To our knowledge, none of the cited occlusion-aware tracking works formulates the joint obstacle-and-visibility problem as a binary polynomial optimization; our search covered IEEE Xplore, arXiv, and Scopus for the terms $\{\text{QUBO, HUBO, Ising, annealing, QAOA}\} \times \{\text{occlusion, visibility, target tracking, UAV path planning}\}$.

III. PROBLEM AND FORMULATION

A. Time Convention and Variables

Let $N \times N$ be the grid, V its cell set ($|V| = N^2$), $O \subset V$ the obstacle cells, and L the number of *time layers*, indexed $t = 0, \dots, L-1$. A trajectory is a cell sequence $(v_0, \dots, v_{L-1})$. The binary variable $x_{t,v} = 1$ iff the vehicle occupies v at layer t . The variable count is LN^2 ; for our main instance $L=15$, $N=8$, giving $15 \times 64 = 960$ variables.

B. Motion Relation

We write $u \sim v$ for the permitted-transition relation, defined as the four cardinal neighbours of v *together with v itself*:

$$\mathcal{N}(v) = \{v\} \cup \{4\text{-cardinal neighbours of } v\}. \quad (1)$$

The self-loop permits hovering, which is physically meaningful for a quadrotor. We note explicitly that the self-loop is *not* required for the cubic term of Section III-E to be active: (4) fires on any two-step walk through distinct connected buffer cells.

C. Hamiltonian

The objective decomposes into structural (hard) and quality (soft) terms:

$$H(x) = \underbrace{H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}}}_{\text{penalty-encoded structural}} + \underbrace{H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}}}_{\text{quality}}. \quad (2)$$

The structural terms are *penalty-encoded*: they are finite penalties, not mathematically hard constraints, which is precisely why infeasible samples occur. They are quadratic: $H_{\text{uniq}} = \lambda_u \sum_t (\sum_v x_{t,v} - 1)^2$ (one-hot occupancy per layer); $H_{\text{start}} = \lambda_s (1 - x_{0,v_0})$; $H_{\text{move}} = \lambda_m \sum_{t=0}^{L-2} \sum_{(u,v): u \notin \mathcal{N}(v)} x_{t,u} x_{t+1,v}$ over ordered cell pairs; and $H_{\text{obs}} = \lambda_o \sum_t \sum_{v \in O} x_{t,v}$. Weights are listed in Table I.

D. Occlusion Term

For cells u, w let $\text{LOS}(u, w)$ be the set of cells strictly between u and w on the Bresenham line (endpoints excluded; ties broken toward increasing row index). Bresenham is a *raster approximation*: it can omit cells a continuous ray clips at a corner, so a supercover or exact ray-cell intersection would be strictly more conservative. With a *static* target cell u_{tgt} ,

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_{t=0}^{L-1} \sum_{u \in V} x_{t,u} |\text{LOS}(u, u_{\text{tgt}}) \cap O|. \quad (3)$$

Note this counts *all* intersected obstacle cells rather than testing whether any obstacle blocks the line. Physical line of sight is blocked by the first obstacle; we therefore describe (3) as an *occlusion-depth surrogate*, not a binary visibility test. A binary alternative $\mathbf{1}\{\text{LOS}(u, u_{\text{tgt}}) \cap O \neq \emptyset\}$ is equally admissible; we retain the count because it grades cells by how deeply occluded they are. For a static target this term is linear in x .

E. Sustained-Proximity Term (Cubic)

Let $B \subset V \setminus O$ be the *buffer cells*: non-obstacle cells within Chebyshev distance 1 of some obstacle. Linger in the buffer region is undesirable because small disturbances may cause collision, but transient passage must remain permitted. We therefore penalize *three consecutive layers spent inside a connected chain of buffer cells*:

$$H_{\text{prox}} = \lambda_p \eta \sum_{t=1}^{L-2} \sum_{u \in B} \sum_{v \in B \cap \mathcal{N}(u)} \sum_{w \in B \cap \mathcal{N}(v)} x_{t-1,u} x_{t,v} x_{t+1,w}. \quad (4)$$

Here $\eta = 3$ is a fixed per-triple factor (one unit per buffer cell in the triple), so every cubic coefficient equals $\lambda_p \eta = 5 \times 3 = 15$. The three cells u, v, w need *not* coincide; they must merely be buffer cells forming a legal two-step walk. This is essential: because $\mathcal{N}$ includes the self-loop, the same-cell case $u=v=w$ is included as a special case, but the term is dominated by genuinely distinct triples. On our instance (4) yields 3,172 cubic monomials, of which 312 (9.8%) are same-cell and 2,860 (90.2%) involve at least two distinct cells. The term is therefore nonzero on ordinary feasible trajectories that traverse the buffer region, and it is the sole source of degree-three structure in H .

Two goal terms remain. H_{goal} is a time-weighted Euclidean distance-to-target cost,

$$H_{\text{goal}} = \lambda_g \sum_{t=0}^{L-1} \sum_{v \in V} \frac{t+1}{L} \|p_v - p_{u_{\text{tgt}}}\|_2 x_{t,v}, \quad (5)$$

which pulls the trajectory toward the target with weight increasing in t ; and $H_{\text{term}} = \lambda_t \sum_{v \neq u_{\text{tgt}}} x_{L-1,v}$ penalizes every non-target cell at the final layer. Both are linear. The *soft cost* reported throughout is $H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}}$ evaluated on the decoded trajectory.

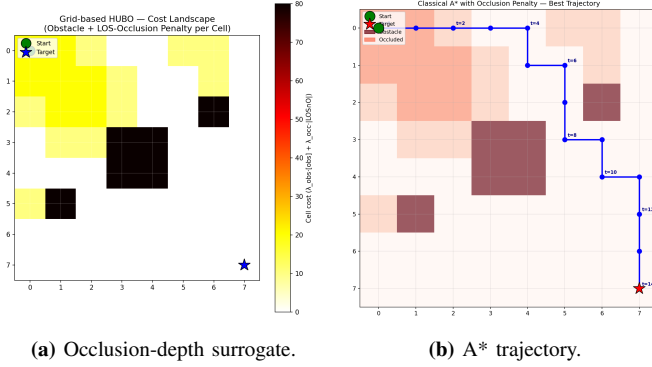


Fig. 1: Output of the instance builder and A* baseline (task2_grid_hubo.py, astar_baseline.py). (a) Per-cell cost surface combining obstacle penalty and $|\text{LOS}(u, u_{\text{tgt}}) \cap O|$; dark cells are obstacles. (b) A* trajectory on the same instance: structurally feasible, goal-reaching, visibility rate 80%.

F. Reduction to QUBO

Quadratic-only samplers cannot evaluate (4). Rosenberg reduction [10], [11] introduces, per cubic product, a *product (reduction) variable* $y_{t,u,v}$ constrained by penalty terms to equal $x_{t-1,u}x_{t,v}$, then substitutes $y_{t,u,v}x_{t+1,w}$. Applying `dimod.make_quadratic` [14] to our instance:

Original HUBO variables	960
Cubic monomials	3,172
Product (reduction) variables	582 (+60.6%)
Reduced QUBO variables	1,542
Quadratic interactions	88,470

This reduced QUBO—not the HUBO—is what a quadratic-only sampler receives. Section VI discusses why the reduction penalty strength matters and why we cannot presently attribute solver behaviour to the reduction alone.

IV. EVALUATION PROTOCOL

A. What “Feasibility-Aware” Means

We call an evaluation protocol *feasibility-aware* when it decouples hard-constraint satisfaction from objective quality at every stage, rather than summarizing a solver by a single aggregate energy. The motivation is concrete: because hard constraints are folded into the objective as penalties, a penalty-based solver can lower its reported energy by *violating* a constraint. A single energy number therefore conflates “found a good trajectory” with “found a low-energy bitstring that is not a trajectory at all.” Any feasibility test that thresholds on energy inherits the same defect, since the threshold depends on the very penalty weights being validated. Concretely, feasibility-aware evaluation requires: separate success metrics (Section IV-B); constraint checks by direct inspection of decoded trajectories; a common external evaluator applied to every solver’s output; matched sampling budgets; and a penalty-weight sweep in place of a single operating point.

B. Three Distinct Success Metrics

Conflating constraint satisfaction with task success overstates performance. We report three metrics separately:

- **Structural feasibility:** one-hot occupancy at every layer; correct start cell v_0 ; every consecutive pair satisfying $v_{t+1} \in \mathcal{N}(v_t)$; no obstacle occupancy. All four are checked by direct inspection of the decoded trajectory, never by an energy threshold.
- **Goal arrival:** $v_{L-1} = u_{\text{tgt}}$.
- **Full-task success:** structural feasibility and goal arrival together.

C. Common External Evaluator

All reported hard penalties and soft costs are recomputed from the original HUBO terms after decoding, regardless of a solver’s internal objective. A* minimizes a per-cell surrogate and terminates on reaching the goal; the stochastic solvers minimize H directly. Their outputs are nonetheless scored by the same external evaluator, so the soft costs in Table II are directly comparable.

D. Solvers

(a) **A*** [9]: classical search with per-cell cost $1 + \lambda_{\text{occ}}|\text{LOS} \cap O| + \lambda_p \mathbf{1}[v \in B]$ and Manhattan heuristic. This is an approximate surrogate for the HUBO soft objective: its buffer penalty is per-cell rather than three-layer, and goal reaching is enforced by termination rather than H_{term} .

(b) **Constrained trajectory SA:** classical SA whose state is a trajectory and whose proposals preserve $\mathcal{N}$ -connectivity, the start cell, and obstacle avoidance. Structural feasibility is therefore guaranteed *by construction*, not discovered by search; we label it accordingly and do not present its feasibility rate as evidence of search quality. It evaluates H natively, computing each monomial as a literal product of indicators regardless of degree.

(c) **neal** [13]: a *classical* CPU simulated-annealing sampler from D-Wave’s Ocean stack. Its API (`sample_qubo`, `sample_ising`, `sample`) is quadratic-only; there is no higher-order method. We apply it through `dimod.HigherOrderComposite` [14], which performs the reduction of Section III-F before sampling. It therefore searches the 1,542-variable reduced QUBO.

(d) **Qiskit QAOA** [12], [15]: gate-based variational optimization on a state-vector simulator. Qiskit’s `QuadraticProgram` accepts only linear and quadratic terms.

E. Instances

Tier 1: $N=8$, $L=15$, start $(0, 0)$, target $(7, 7)$, six obstacles, $|B| = 24$ buffer cells. This yields $15 \times 64 = 960$ binary variables and 87,684 monomials, of which 960 are linear, 83,552 quadratic, and 3,172 cubic; maximum coefficient magnitudes by degree are 1975/2000/15. All counts in this paper are regenerated from the instance builder at the cited repository and can be reproduced by executing it. Solvers: A*, constrained SA, **neal**. **Tier 2:** $N=2$, $L=2$, 8 variables, no obstacles—hence $B = \emptyset$, $H_{\text{prox}} \equiv 0$, and H_{occ} identically zero. This instance is *already quadratic*; it exercises the QAOA encoding path only and tests neither occlusion nor higher-order structure.

TABLE I: Objective and solver parameters.

Symbol	Role	Value
$\lambda_u, \lambda_s, \lambda_m$	swept structural penalties	1000
λ_o	obstacle penalty (<i>not</i> swept)	80
λ_{occ}	occlusion-depth weight	10
λ_p	sustained-proximity weight	5
λ_g	distance-to-target weight	8
λ_t	terminal penalty	50
η	per-triple factor in H_{prox}	3
max coef by degree	1/2/3	1975/2000/15
penalty_strength	dimod reduction	1.0 (default)
discard_unsatisfied	dimod	False (default)
SA sweeps / reads	constrained SA	500 / 50
neal sweeps / reads	Tier 1 main run	3000 / 50
neal reads	penalty sweep	15
seed	all stochastic runs	42

TABLE II: Tier 1 results, 8×8 , $L=15$. Counts are k/n with Wilson 95% intervals. *neal* operates on the reduced 1,542-variable QUBO; constrained SA is feasible by construction. Soft cost: lower is better, recomputed by a common evaluator.

	A*	Constr. SA	neal
Structural feasibility	1/1	50/50 [‡]	11/50 [12.8, 35.2]%
Goal arrival	1/1	7/50 [7.0, 26.2]%	4/50 [3.2, 18.8]%
Best soft cost	317	303	496
Visibility rate	80%	— [†]	— [†]

[‡]Guaranteed by the proposal operator, not discovered by search. [†]Per-step trajectories were not retained; see Section VI.

V. RESULTS

A. Tier 1

Table II reports all metrics as raw counts with Wilson 95% intervals. A* returned a structurally feasible, goal-reaching trajectory for this instance with soft cost 317. The constrained SA attains the best soft cost observed (303) and is feasible by construction, but reaches the goal in only 7/50 samples. *neal* on the reduced QUBO attains 11/50 structural feasibility and 4/50 goal arrival, with best feasible soft cost 496.

We report best soft cost, but note that “best of n ” favours high-variance methods. Median and IQR over full-task-successful samples would be more informative; because the per-sample feasibility/arrival join was not retained, we cannot compute them here and do not report them.

B. Penalty-Weight Sweep

Fig. 2 reports structural feasibility against the structural penalty λ . *This is a separate run using 15 reads per weight, not the 50-read run of Table II*; the two therefore differ at $\lambda=1000$ ($3/15 = 20\%$ versus $11/50 = 22\%$), which is well within sampling noise. Counts are 0/15 at $\lambda \in \{50, 100, 300\}$, 2/15 at 600, 3/15 at 1000, and 9/15 at 2000.

The qualitative pattern—feasibility rising steeply only once λ exceeds the objective scale by roughly an order of magnitude—is robust to the small sample size, but the individual rates carry wide intervals (0/15 has a 95% upper bound of 20.4%) and should not be read as precise.

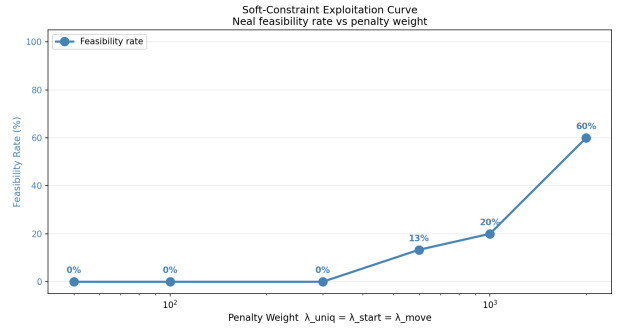


Fig. 2: Structural feasibility of *neal* on the reduced QUBO versus structural penalty weight λ (log scale), as produced by `task3_polished_v3_1.py`. Each point is $n=15$ reads; raw counts are 0/15, 0/15, 0/15, 2/15, 3/15, 9/15. Wilson 95% intervals are wide (e.g. 0/15 has upper bound 20.4%) and are given in the text; no regime boundary is drawn, as we have no principled threshold to justify one.

C. Tier 2 Encoding Check

On the 8-variable obstacle-free instance, exhaustive enumeration gives optimum $E^* = -118.17$. Each of constrained SA, *neal*, and Qiskit QAOA (SPSA, reps= 1, maxiter= 25) produced at least one sample attaining this exact energy; run-times were 0.14 s, 0.02 s, and 55 s. We did *not* record optimal-sampling probability, QAOA expected energy, approximation ratio, shot count, optimizer trace, or exact Qiskit component versions, all of which are required before any statement about QAOA performance. Because this instance contains neither obstacles nor buffer cells, it exercises neither H_{occ} nor H_{prox} : it validates the encoding path into `QuadraticProgram` and nothing more. We report only best observed energy; expected energy, optimal-sampling probability, and approximation ratio were not recorded, and a strengthened QAOA study is required before any statement about QAOA *performance* is warranted.

VI. LIMITATIONS AND CONFOUNDS

We state these explicitly because several bear directly on how the results may be interpreted.

Reduction penalty strength. `HigherOrderComposite` was invoked at its default `penalty_strength= 1.0` (`dimod.make_quadratic` itself takes `strength` as a required argument; the default arises from the composite). The correct scale for this penalty is *not* the largest coefficient in H (2000, which comes from expanding the one-hot term), but the energy benefit obtainable by violating an individual product constraint, which is bounded by the coefficients of terms containing that product. Each cubic coefficient is $\lambda_p \eta = 15$, but a single product variable $y_{t,u,v}$ may be shared across several cubic terms, so the correct bound is the *sum of incident coefficient magnitudes* over all terms using that product, not one coefficient of 15. We did not compute this per-product sum, so we cannot state how far short a strength of 1.0 falls; we can only say it is demonstrably not always sufficient, since reductions are violated in practice. `dimod` documentation notes that insufficient strength can change the minimizers after reduction. With `discard_unsatisfied=False`, sam-

ples violating a reduction were retained. Auditing the full 50-read run with `keep_penalty_variables=True` gives 44/50 product-consistent reads (12% violation). Note also that `HigherOrderComposite` returns *original polynomial* energies excluding reduction penalties, so native-HUBO and reduced-QUBO energies are distinct quantities; we report the former. *Consequently we do not claim that quadratization causes `neal`'s lower feasibility.* Competing explanations—weak reduction strength, sampler schedule, dense reduced topology, and unequal budget—are not separated by the present experiments. The required ablation is: `neal` on (i) the instance with $\lambda_p=0$ (no cubic terms), (ii) the reduced QUBO at several reduction strengths, and (iii) bit-level SA on the same QUBO at matched schedule, with product-consistency reported per sample.

Unequal computational budget. Fifty reads allocate ≈ 2390 s to constrained SA and ≈ 13 s to `neal`. Equal read counts control one axis only; equal wall-clock and equal objective-evaluation comparisons are needed.

Search-space asymmetry. Constrained SA searches only feasible trajectories; `neal` searches unconstrained bitstrings. Their feasibility rates are therefore not comparable as search outcomes.

Full-task success not separable. Structural feasibility and goal arrival are distinct events, so full-task success is their intersection. Our retained logs record the two rates (11/50 and 4/50 for `neal`; 50/50 and 7/50 for constrained SA) but not the per-sample join, so the intersection cannot be recovered post hoc. Soft-cost statistics should be conditioned on full-task success rather than structural feasibility alone; the present “best soft cost” figures are conditioned on feasibility only and are therefore optimistic.

Missing visibility data. Full trajectories for the stochastic solvers were not retained, so visibility rate $\frac{1}{L} \sum_t \mathbf{1}\{\text{LOS}(v_t, u_{\text{tgt}}) \cap O = \emptyset\}$ is reported only for A*. This must be rerun.

Horizon is exactly tight. With start (0,0), target (7,7) and $L=15$ layers there are exactly 14 transitions, equal to the Manhattan distance. Every goal-reaching trajectory is therefore a shortest monotone path: no hovering, no backtracking, and no detour to improve visibility is possible while still arriving. This substantially weakens the occlusion/path-length trade-off the formulation is meant to expose, and means the reported soft-cost differences reflect choice *among* shortest paths only. A longer horizon ($L=18-20$), or early arrival followed by self-loops at the target, is required before the trade-off can be studied. We regard this as the most consequential design flaw in the present instance.

Model abstraction and task mismatch. The model is a 2D grid planner without vehicle dynamics; “UAV” should be read as motivation. Moreover the objective drives the vehicle *into* the target cell, whereas target *tracking* normally requires a standoff band $s_{\min} \leq \|p_t - r_t\| \leq s_{\max}$ about a moving target. The instance studied here is therefore a static goal-navigation problem with visibility preference, not a tracking problem; restoring a standoff band and a known target trajectory is

required to model tracking proper.

Static target only. We make no claim about the degree of a moving-target formulation. With target occupancy $y_{t,w}$ and precomputed blockage $c_{u,w}$, the natural moving-target visibility term $\sum_t \sum_{u,w} c_{u,w} x_{t,u} y_{t,w}$ is *quadratic*, not cubic; a higher-order moving-target model would require a third decision factor that we have not formulated.

VII. DISCUSSION AND CONCLUSION

On this static 8×8 instance A* is decisively the strongest method: A* returned a goal-reaching trajectory for this instance, which neither stochastic method does reliably, at a per-call cost of 10 ms against tens of seconds (SA) and fractions of a second (`neal`) per *sample*. We deliberately avoid quoting a speed-up ratio: a single deterministic A* call and a stochastic per-sample cost are not commensurable, and we did not record machine specification, repeated timings, variance, or whether model construction and quadratization are included. Equal-wall-clock and equal-objective-evaluation comparisons remain to be done. We state this plainly because it supports rather than undercuts the paper’s position: the value of a binary-polynomial formulation is not that it beats shortest-path search on a small static grid, but that it supplies a precisely specified, reproducible benchmark instance with genuine higher-order structure for solvers that consume such models.

The feasibility-aware protocol is the transferable contribution. Separating structural feasibility from goal arrival, verifying constraints by inspection, reporting k/n with intervals, and sweeping the penalty weight together prevent a family of comparisons that would otherwise look favourable and mean little.

Future work, in order: rerun with full trajectory logging and the reduction ablation of Section VI; add equal-wall-clock comparisons; strengthen the QAOA study to report approximation ratio and optimal-sampling probability on an instance that actually contains occlusion; and formulate a moving-target model whose degree is derived rather than asserted.

REFERENCES

- [1] J. Chen, T. Liu, and S. Shen, “Tracking a moving target in cluttered environments using a quadrotor,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2016, pp. 446–453, doi:10.1109/IROS.2016.7759092.
- [2] K. Hausman, G. Kahn, S. Patil, J. Müller, K. Goldberg, P. Abbeel, and G. S. Sukhatme, “Occlusion-aware multi-robot 3D tracking,” in *Proc. IEEE/RSJ IROS*, 2016, pp. 1863–1870, doi:10.1109/IROS.2016.7759296.
- [3] B. Penin, P. Robuffo Giordano, and F. Chaumette, “Vision-based reactive planning for aggressive target tracking while avoiding collisions and occlusions,” *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3725–3732, 2018, doi:10.1109/LRA.2018.2856526.
- [4] D. Falanga, P. Foehn, P. Lu, and D. Scaramuzza, “PAMPC: Perception-aware model predictive control for quadrotors,” in *Proc. IEEE/RSJ IROS*, 2018, pp. 1–8, doi:10.1109/IROS.2018.8593739.
- [5] J. Wang, Y. X. Wu, Y.-Q. Chen, and S. Ju, “Multi-UAVs collaborative tracking of moving target with maximized visibility in urban environment,” *J. Franklin Inst.*, vol. 359, no. 11, pp. 5512–5532, 2022, doi:10.1016/j.jfranklin.2022.05.004.
- [6] H. Masnavi, V. K. Adajania, K. Kruusamäe, and A. K. Singh, “Real-time multi-convex model predictive control for occlusion-free target tracking with quadrotors,” *IEEE Access*, vol. 10, pp. 29009–29031, 2022, doi:10.1109/ACCESS.2022.3157977.

- [7] N. Innan, M. Kashif, A. Marchisio, Y. S. Gan, F. Barbaresco, and M. Shafique, “QUAV: Quantum-assisted path planning and optimization for UAV navigation with obstacle avoidance,” in *Proc. IEEE Int. Conf. Quantum Artif. Intell. (QAI)*, 2025, pp. 208–215, doi:10.1109/QAI63978.2025.00040.
- [8] E. Osaba, P. Miranda-Rodriguez, A. Oikonomakis, M. Petrič, A. Ruiz, S. Bock, and M.-A. Kourtis, “Solving drone routing problems with quantum computing: A hybrid approach combining quantum annealing and gate-based paradigms,” in *Proc. IEEE Congr. Evol. Comput. (CEC)*, 2025, pp. 1–8, doi:10.1109/CEC65147.2025.11042978.
- [9] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Trans. Syst. Sci. Cybern.*, vol. 4, no. 2, pp. 100–107, 1968, doi:10.1109/TSSC.1968.300136.
- [10] I. G. Rosenberg, “Reduction of bivalent maximization to the quadratic case,” *Cahiers du Centre d’Études de Recherche Opérationnelle*, vol. 17, pp. 71–74, 1975.
- [11] F. Glover, G. Kochenberger, and Y. Du, “Quantum bridge analytics I: A tutorial on formulating and using QUBO models,” *4OR*, vol. 17, pp. 335–371, 2019, doi:10.1007/s10288-019-00424-y.
- [12] E. Farhi, J. Goldstone, and S. Gutmann, “A quantum approximate optimization algorithm,” arXiv:1411.4028, 2014, doi:10.48550/arXiv.1411.4028.
- [13] D-Wave Systems, *dwave-neal* simulated annealing sampler [software], v0.6.0. <https://github.com/dwavesystems/dwave-neal>.
- [14] D-Wave Systems, *dimod* [software], v0.12.22, incl. `HigherOrderComposite` and `make_quadratic`. <https://github.com/dwavesystems/dimod>.
- [15] Qiskit contributors, *Qiskit*, *Qiskit Optimization*, *Qiskit Algorithms*, and *Qiskit Aer* [software]. Exact component versions were not recorded for the runs reported here; this is a reproducibility gap noted in Section VI. <https://qiskit.org>.
- [16] S. H. James and R. N. Raheb, “Path planning for critical ATM/UTM areas,” in *Proc. IEEE/AIAA 38th Digital Avionics Systems Conf. (DASC)*, San Diego, CA, USA, 2019, pp. 1–6, doi:10.1109/DASC43569.2019.9081662.
- [17] E. Davies and P. Kalidindi, “Quantum algorithms for drone mission planning,” arXiv:2409.18631, 2024; published as *Proc. SPIE 13202, Quantum Technologies for Defence and Security*, art. 1320209, 2024, doi:10.1117/12.3036340.
- [18] X. Yu and C. W. Chen, “HQC-NBV: A hybrid quantum-classical view planning approach,” arXiv:2505.05212, 2025.
- [19] T. Gerlach, L. K. Lee, F. Barbaresco, and N. Piatkowski, “Hybrid quantum-classical multi-agent pathfinding,” in *Proc. 42nd Int. Conf. Machine Learning (ICML)*, PMLR vol. 267, pp. 19161–19171, 2025.